Prayas JEE AIR (2025)

WEEKLY TEST 26

DURATION : 180 Minutes | **DATE : 04/01/2025** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (2)
3. (4)
4. (1)
5. (2)
6. (4)
7. (3)
8. (1)
9. (3)
10. (2)
11. (2)
12. (3)
13. (3)
14. (2)
15. (4)
16. (2)
17. (2)
18. (4)
19. (3)
20. (2)
21. (75)
22. (100)
23. (100)
24. (10)
25. (40)

CHEMISTRY

26. (2)
27. (1)
28. (3)
29. (1)
30. (4)
31. (3)
32. (3)
33. (4)
34. (4)
35. (3)
36. (1)
37. (3)
38. (1)
39. (2)
40. (2)
41. (1)
42. (2)
43. (3)
44. (3)
45. (2)
46. (0)
47. (82)
48. (57)
49. (4)
50. (104)

MATHEMATICS

51. (2)
52. (3)
53. (1)
54. (1)
55. (2)
56. (3)
57. (1)
58. (2)
59. (2)
60. (1)
61 (1)
62. (3)
63. (4)
64. (3)
65. (1)
66. (2)
67. (2)
68. (2)
69. (4)
70. (1)
71. (6)
72. (3)
73. (12)
74. (2)
75. (1)

SECTION-I (PHYSICS)

1. **(2)**

For upward motion,

$$g_{eff} = \frac{mg + F_a}{m} = \frac{50+10}{5} = 12 \text{ m/s}^2$$

$$h = \frac{u_y^2}{2g_{eff}} = \frac{u^2}{24}$$

$$\Rightarrow u = \sqrt{24h}$$

$$t_a = \frac{u_y}{g_{eff}} = \frac{u}{12} = \frac{\sqrt{24h}}{12} = \sqrt{\frac{h}{6}}$$

For downward motion,

$$g_{eff} = \frac{mg - F_a}{m} = \frac{50-10}{5} = 8 \text{ m/s}^2$$

$$h = ut + \frac{1}{2} g_{eff} t^2$$

$$\Rightarrow h = \frac{1}{2} \times 8 \times t_d^2$$

$$\Rightarrow t_d = \sqrt{\frac{h}{4}}$$

$$\frac{t_a}{t_d} = \frac{\sqrt{4}}{\sqrt{6}} = \sqrt{\frac{2}{3}}$$

2. **(2)**

$P_i = Nmvt$ and $P_f = -Nmvt$

Where N is the number of balls that strike the wall

$$\Delta P = P_f - P_i = -2Nmvt = -200Nmv\hat{i}$$

$$F_{\text{Total}} = \frac{\Delta P}{\Delta t} = -\frac{200mvt}{t}$$

$$|F| = \frac{200mv}{t}$$

3. **(4)**

$x = 3t^2 + 5$

$v = 6t \Rightarrow \Delta W = \Delta K$

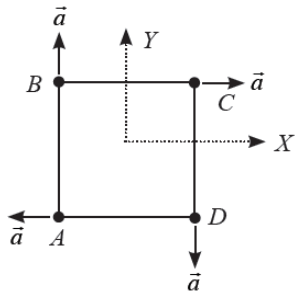
$$= \frac{1}{2}(2)(30)^2 - \frac{1}{2}2(0)^2$$

$= 900\text{J}$

4. **(1)**

$$\vec{a}_{cm} = \frac{m_A\vec{a}_A + m_B\vec{a}_B + m_C\vec{a}_C + m_D\vec{a}_D}{m_A + m_B + m_C + m_D}$$

$$\vec{a}_{cm} = \frac{-ma\hat{i} + 2ma\hat{j} + 3ma\hat{i} - 4ma\hat{j}}{10m}$$



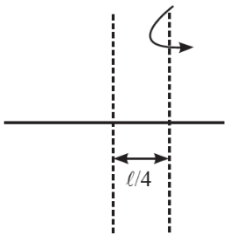
$$\vec{a}_{cm} = \frac{a}{5}\hat{i} - \frac{a}{5}\hat{j} = \frac{a}{5}\left(\hat{i} - \hat{j}\right)$$

5. **(2)**

Moment of inertia

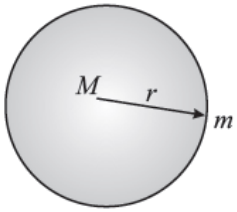
$$I = \frac{M^2}{12} + \frac{M^2}{16} = \frac{7M^2}{48}$$

Comparing with $I = Mk^2$ [k = Radius of inration]



6. **(4)**

Let total mass included in a sphere of radius r be M. For a particle of mass m,



$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\Rightarrow \frac{GMm}{r} = 2K$$

$$\Rightarrow M = \frac{2Kr}{GM}$$

$$\therefore dM = \frac{2Kdr}{GM}$$

$$\Rightarrow \left(4\pi r^2 dr\right)\rho = \frac{2Kdr}{GM}$$

$$\Rightarrow \rho = \frac{K}{2\pi r^2 GM}$$

$$\therefore n = \frac{\rho}{m} = \frac{K}{2\pi r^2 m^2 G}$$

7. **(3)**

By using Hooke's Law

$F \alpha \Delta L$

$$\frac{F_1}{F_2}=\frac{\Delta L_1}{\Delta L_2}$$

$$\Rightarrow \frac{10}{20}=\frac{L_1-L}{L_2-L}$$

$$\Rightarrow L = 2L_1 - L_2$$

8. **(1)**

$$A_A\ v_A = A_B\ v_B$$

$$\Rightarrow v_A = \frac{25\times10^{-6}\times0.6}{1.5\times10^{-4}} = 0.1\text{m/s}$$

$$P_A + \frac{1}{2}\rho v_A^2 = P_B + \frac{1}{2}\rho v_B^2$$

$$\Rightarrow P_A - P_B = \frac{1}{2}\rho\left(v_B^2 - v_A^2\right)$$

$$\Rightarrow P_A - P_B = \frac{1}{2}\times1000\times\left(0.6^2 - 0.1^2\right) = 175\,\text{Pa}$$

9. **(3)**

Type of gases	Degree of freedom		
	Translational	Rotational	Vibrational
Monoatomic	3	0	0
Rigid diatomic	3	2	0
Non-rigid diatomic	3	2	1
Polyatomic	3	3	More than1

10. **(2)**

Second pendulum has time period equal to 2 second. It take 1 second to move between extreme position.

11. **(2)**

$$\Delta U = \frac{4q^2}{4\pi\varepsilon_0}\left[\frac{1}{3d/2} - \frac{1}{d/2}\right]$$

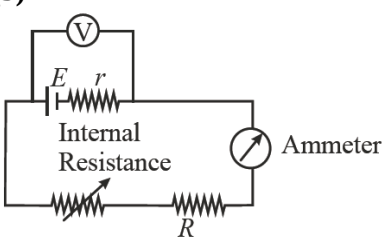
$$\Delta U = \frac{4q^2}{3\pi\varepsilon_0 d}$$

12. **(3)**

$$E = \frac{q_1 - q_2}{2A\varepsilon_0}\ \text{(between plates)}$$

$$V = Ed = \frac{(q_1 - q_2)d}{2A\varepsilon_0} = \frac{q_1 - q_2}{2C}\left(QC = \frac{A\varepsilon_0}{d}\right)$$

13. **(3)**



$V = E - Ir$

When $V = V_0 = 0 \Rightarrow 0 = E - Ir$

$\therefore E = Ir$

When I = 0, V = E = 1.5 V

$\therefore r = 1.5\ \Omega$

14. **(2)**

Number of turns = N

Inner radius = a, outer radius = b,

Consider an elementary portion of radius r and thickness dr.

Number of turns in elementary portion,

$$dN = \frac{N}{2\pi(b-a)}\cdot 2\pi dr = \frac{N}{(b-a)}dr$$

$$\therefore (\text{dB})\text{at centre} = \frac{\mu_0 I dN}{2r} \Rightarrow B = \int_a^b \frac{\mu_0 I}{2r}\times\left(\frac{N}{b-a}\right)$$

$$\Rightarrow B = \frac{\mu_0 IN}{2(b-a)}\log_e\left(\frac{b}{a}\right)$$

15. **(4)**

$$v = \frac{E}{B}$$

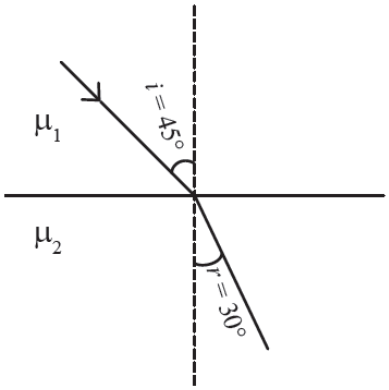
$$r = \frac{mv}{qB}$$

$$\Rightarrow m = \frac{rqB}{v} = \frac{rqB^2}{E}$$

$$\Rightarrow m = \frac{0.5}{100}\times1.6\times10^{-19}\times\frac{(0.5)^2}{100}$$

$$\Rightarrow m = 2\times10^{-24}\,\text{kg}$$

16. **(2)**



Using snell's law, $\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}, \frac{\sin 45°}{\sin 30°} = \frac{\mu_2}{\mu_1}$

$$\frac{\mu_1}{\mu_2} = \frac{1}{\sqrt{2}}$$

$\therefore$ Refractive Index $(\mu) \propto \frac{1}{\text{wavelength}(\lambda)}$

$$\therefore \frac{\mu_1}{\mu_2} = \frac{\lambda_2}{\lambda_1} = \frac{1}{\sqrt{2}}$$

$$\lambda_2 = \frac{\lambda_1}{\sqrt{2}}$$

Frequency will same in both medium.

17. (2)

From the question, $\frac{A_1}{A_2}=\frac{2}{1}$

$I \propto (\text{Amplitude})^2$

$\frac{I_{max}}{I_{min}}=\left(\frac{A_1+A_2}{A_1-A_2}\right)^2=\frac{9}{1}=9:1$

18. (4)

Electric field at $t = 0$ & $(x,y,z)=\left(0,0,\frac{\pi}{k}\right)$

$\vec{E}=-\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)E_0$ *and* $\vec{F}=\vec{E}q$

Due to electric field $\vec{F}=\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)q$

Hence, net force is antiparallel to $\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

19. (3)

$E = E_1 + E_2$

$13.6(2)^2\left(\frac{1}{1^2}-\frac{1}{n^2}\right)=\frac{1240}{108.5}+\frac{1240}{30.4}$

$\left(1-\frac{1}{n^2}\right)=\frac{52.22}{54.4}$

$\frac{1}{n^2}=\frac{2.18}{54.4} \Rightarrow n^2 = 25$

$\Rightarrow n = 5$

20. (2)

Energy = (Binding Energy)$_{\text{Reactant}}$ – (Binding Energy)$_{\text{Product}}$

= 20 × 8.03 – (8 × 7.07 + 12 × 7.86)

= 160.6 – (56.56 + 94.32) = 160.6 – 150.88 = 9.72 MeV

Hence, 9.72 MeV is released

21. (75)

Given :

$V_{air} = 300\,\text{m/s}$, $\rho_{gas} = 2\rho\,\text{air}$

Using $V=\sqrt{\frac{B}{\rho}}$

$\frac{V_{gas}}{V_{air}}=\frac{\sqrt{\frac{B}{2\rho_{air}}}}{\sqrt{\frac{B}{\rho_{air}}}} \Rightarrow V_{gas}=\frac{V_{air}}{\sqrt{2}}=\frac{300}{\sqrt{2}}=150\sqrt{2}\,\text{m/s}$

And f_{nth} harmonic = $\frac{nv}{2L}$ (in open organ pipe)

(L =1 metre given)

$\therefore f_{2nd}\text{harmonic} - f_{\text{fundamental}}=\frac{2v}{2\times1}-\frac{v}{2\times1}=\frac{v}{2}$

$\therefore f_{2n}\text{harmonic} - f_{\text{fundamental}}$

$=\frac{150\sqrt{2}}{2}=\frac{150}{\sqrt{2}}=75\sqrt{2}$ Hz

22. (100)

Volume of big drop = 1000 × volume of each small drop

$\frac{4}{3}\pi R^3 = 1000\times\frac{4}{3}\pi r^3 \Rightarrow R = 10\,r$

$\because V = \frac{kq}{r}$ and $V' = \frac{kq}{R}\times1000$

Total charge on one small droplet is q and on the big drop is 1000q.

$\Rightarrow \frac{V'}{V}=\frac{1000r}{R}=\frac{1000}{10}=100$

$\therefore V' = 100\,V$

23. (100)

$X_L = \omega L = (2\pi\times500)\times8.1 = 25.4\Omega$

and $X_C=\frac{1}{\omega C}=\frac{1}{(2\pi\times500)\times\left(12.5\times10^{-6}\right)}$

= 25.4 Ω

As $X_L = X_C$, so resonance will occur and V_R = 100 V

24. (10)

de-Broglie wavelength (λ) is given by

$\lambda=\frac{h}{p}=\frac{h}{\sqrt{2mK}}=\frac{h}{\sqrt{2mqV}}$

$\left(\because p=\sqrt{2mK}\right)$

Substituting the values we get

$\therefore \frac{\lambda_A}{\lambda_B}=\frac{\sqrt{2m_Bq_BV_B}}{\sqrt{2m_Aq_AV_A}}=\sqrt{\frac{4m\times q\times2500}{m\times q\times50}}$

$= 2\sqrt{50}=10\sqrt{2}$

25. (40)

Using the principal of calorimetry

$M_{ice}\,L_f + m_{ice}\,(40-0)\,C_w$

$= m_{stream}\,L_v + m_{stream}\,(100-40)\,C_w$

$\Rightarrow M\,(540) + M\times1\times(100-40)$

$= 200\times80 + 200\times1\times40$

$\Rightarrow 600\,M = 24000$

$\Rightarrow M = 40$ g

SECTION-II (CHEMISTRY)

26. (2)

$$[N_2O_5] = \frac{\text{Rate}}{K} = \frac{2.4\times10^{-5}}{6\times10^{-5}} = 0.4\,\text{mol}\,L^{-1}$$

27. (1)

Apply law of equivalence:

$m_{cq.}$ of $HCl = m_{cq}$ of Na_2CO_3

$25\times N = 30\times0.1\times2$

$$N_{HCl} = \frac{30\times0.2}{25} = \frac{6}{5}\times0.2 = \frac{1.2}{5}$$

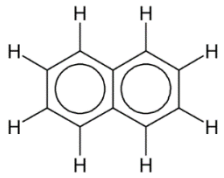
For the 2nd titration

$m_{eq.}$ of $HCl = m_{eq.}$ of $NaOH$

$$\frac{1.2}{5}\times V_{HCl} = 30\times0.2$$

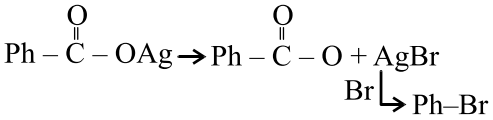
$$V_{HCl} = \frac{6\times5}{1.2} = \frac{30}{1.2} = 25\,\text{mL}$$

28. (3)

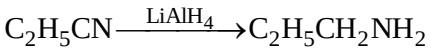


29. (1)

This is Hundsdicker reaction.



30. (4)



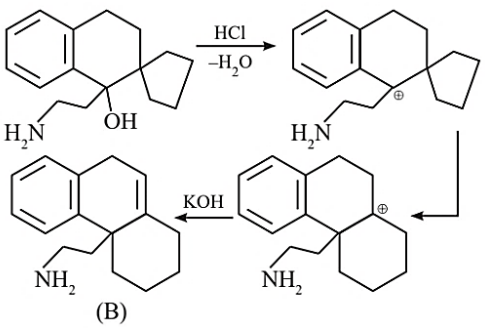
31. (3)

On moving left to right along a period, atomic radius decreases

32. (3)

In this reaction, firstly carbocation will be formed which will undergo ring expansion to form the more stable carbocation and further in the presence of KOH, the product (B) is formed

The reaction sequence is given below



33. (4)

The correct matches for the chemicals to their corresponding use/preparation or constituent are:

Alcoholic potassium hydroxide (Alc.KOH): used for b-elimination

$Pd/BaSO_4$: Lindlar's Catalyst

BHC (Benzene hexa chloride): Obtained by addition reaction

Graphite: Electrodes in batteries.

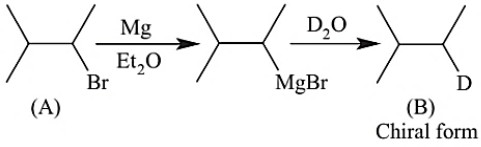
34. (4)

Acc. to Fajan's rule a compound is said to be covalent having smaller cation, larger anion and having high charge density.

(1), (2) and (3) are factors which affect the percent covalent character of the ionic bond according to Fajan's rule

35. (3)

The reaction occurs as:



36. (1)

Amylose is a linear chain polymer of α-D-glucose while amylopectine is a branched chain polymer of α-D-glucose.

37. (3)

$5[M_A + 2M_B] = 125$

$M_A + 2M_B = 25$ ...(i)

$2M_A + 2M_B = 30$ ...(ii)

from eq. (i) & (ii)

$M_A = 5$

$M_B = 10$

38. (1)

According to question all the complexes are low spin.

Complex	Configuration	No. of unpaired electrons
$[V(CN)_6]^4$	$t_{2g}^3 e_g^o$	3
$[Cr(NH_3)_6]^{2+}$	$t_{2g}^4 e_g^0$	2
$[Ru(NH_3)_6]^{3+}$	$t_{2g}^5 e_g^0$	1
$[Fe(CN)_6]^4$	$t_{2g}^6 e_g^0$	0

39. (2)

Rate constant depends on temperature as per Arrhenius equation:

$K = Ae^{-Ea/RT}$ $\ln k = \ln A - \dfrac{Ea}{RT}$

As T increases, rate constant increases exponentially.

40. (2)

In option (2), no conjugation will take place.

41. (1)

$2Cu^{+} \rightarrow Cu^{2+} + Cu$

The stability of $Cu^{2+}(aq)$ is more than $Cu^{+}(aq)$ due to the much more negative $\Delta_{hyd}H$ of $Cu^{2+}(aq)$ than $Cu^{+}(aq)$.

42. (2)

For the salt of strong acid and weak base

$$\left[H^{+}\right] = \sqrt{\frac{K_w \times C}{K_b}}$$

$$\left[H^{+}\right] = \sqrt{\frac{10^{-14} \times 2 \times 10^{-2}}{10^{-5}}}$$

$$-\log\left[H^{+}\right] = 6 - \frac{1}{2}\log 20$$

$\therefore \quad pH = 5.35$

43. (3)

Moles of HCl reacting with ammonia = (moles of HCl absorbed) – (moles of NaOH solution required)

$= (20 \times 0.1 \times 10^{-3}) - (15 \times 0.1 \times 10^{-3})$

= moles of NH_3 evolved.

= moles of nitrogen in organic compound

$\therefore$ wt. of nitrogen in org. comp $= 0.5 \times 10^{-3} \times 14$

$= 7 \times 10^{-3}$ g

$$\%wt = \frac{7 \times 10^{-3}}{29.5 \times 10^{-3}} \times 100 = 23.7\%$$

44. (3)

$$x_2 = \frac{n_2}{n_1 + n_2}$$

Mass of solution $= n_1M_1 + n_2M_2$

Volume of solution $= \dfrac{(n_1M_1 + n_2M_2)}{\rho}$

$$M = \frac{n_2}{V} = \frac{n_2\rho}{(n_1M_1 + n_2M_2)} = \frac{x_2\rho}{x_1M_1 + x_2M_2}$$

45. (2)

Attack of nucleophile on carbonyl centres depends upon

(i) Sterric factor

(ii) Electronic factor

$\therefore$ Rate of eacton sould follow the order

(ii) > (iv) > (i) > (iii)

46. (0)

Orbital angular momentum $= \sqrt{l(l+1)}\,\dfrac{h}{2\pi}$

On comparing the formula of orbital angular momentum given above with the value given in question, we get $x = \sqrt{l(l+1)}$

The value of l for s = 0

Therefore, the value of x will be zero.

47. (82)

Boiling point of solvent is 82°C

Boiling point of solution is 83°C

48. (57)

For KCl, $k_1 = 0.14 Sm^{-1}$, $R_1 = 4.19\Omega$

For HCl, $k_2 = x\ Sm^{-1}$, $R_2 = 1.03\ \Omega$

If cell is same, then cell constant does not changes

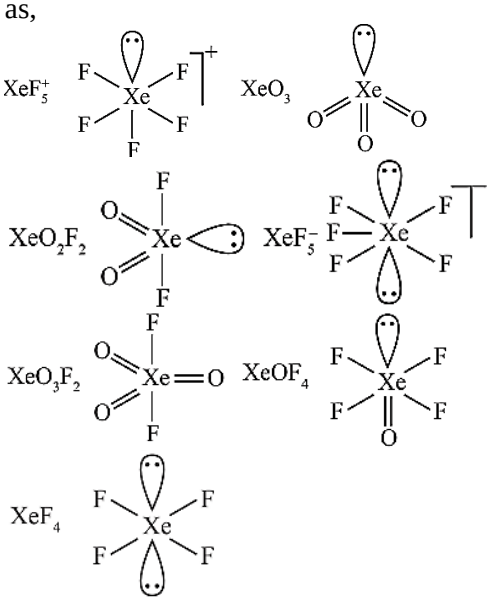
$\therefore \quad \kappa$ = R Constant

$k_1R_1 = k_2R_2 \Rightarrow 0.14 \times 4.19 = 1.03 \times k_2$

$\Rightarrow \quad k_2 = 0.569 = 57 \times 10^{-2}\ Sm^{-1}$

49. (4)

The geometry of all the given species is shown as,



From all the above given structures, it is clear that the species XeF_5^{+}, XeO_3, $XeOF_4$, and XeO_2F_2 carries single lone pair of electrons. So, Answer is 4.

50. (104)

$C_3H_8(g) + 5O_2 \rightarrow 3CO_2 + 4H_2O(l), \Delta H_1$

$= -2220\,kJ/mol$

$C(\text{ graphite }) + O_2(g) \rightarrow CO_2(g), \Delta H_2$

$= -393.5\,kJ/mol, \ (\Delta H_3 = -285.8\,kJ/mol)$

$3C(\text{ Graphite }) + 4H(g) \rightarrow C_3H_8(g)$

$$\Delta H_r(C_3H_8) = \left[3 \times \Delta H_{comb}(C)\right] + \left[4 \times \Delta H_{comb}(H_2)\right] - \left[\Delta H_{comb}(C_3H_8)\right] = -103.7\,kJ/mole$$

SECTION-III (MATHEMATICS)

51. (2)

Let $e^x = t \in (0,\infty)$

Given equation

$t^4 + t^3 - 4t^2 + t + 1 = 0$

$\Rightarrow t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$

$\Rightarrow \left(t^2 + \frac{1}{t^2}\right) + \left(t + \frac{1}{t}\right) - 4 = 0$

Let $t + \frac{1}{t} = \alpha \qquad (\alpha \geq 2)$

$\Rightarrow (\alpha^2 - 2) + \alpha - 4 = 0$

$\Rightarrow \alpha^2 + \alpha - 6 = 0$

$\alpha = -3, 2 (\because \alpha \neq -3) \therefore \alpha = 2 \Rightarrow e^x + e^{-x} = 2$

Here, $x = 0$ only solution

52. (3)

Given, $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix} = \text{adj}(A)$

$\Rightarrow |B| = |\text{adj}(A)| = |A|^2$

$\Rightarrow |B| = 4$

$\Rightarrow 1(8 - 3\alpha) - 3(4 - 3\alpha) + \alpha(\alpha - 2\alpha) = 4$

$\Rightarrow -\alpha^2 + 6a - 8 = 0$

$\Rightarrow \alpha = 2, 4$

Given, $\alpha > 2$

So, $\alpha = 2$ is rejected

$$\begin{bmatrix} 4 & -8 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix} = [-16]_{1\times 1}$$

53. (1)

$\Rightarrow (1 + x + x^2 + x^3)^6$

$= [(1 + x) + x^2 (1 + x)]^6$

$= (1 + x)^6 (1 + x^2)^6$

Coefficient of x^4

$= {}^6C_0 \times {}^6C_2 + {}^6C_2 \times {}^6C_1 + {}^6C_4 \times {}^6C_0$

$= 1 \times 15 + 15 \times 6 + 15 \times 1$

$= 15 + 90 + 15 = 120$

$\therefore$ Coefficient of $x^4 = 120$

54. (1)

$$z = \frac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$$

$$= \frac{i-1}{\frac{1}{2} + i\frac{\sqrt{3}}{2}}$$

$$= \frac{2(i-1)}{1+\sqrt{3}i}$$

$$= \frac{2i-2}{1+\sqrt{3}i} \times \frac{1-\sqrt{3}i}{1-\sqrt{3}i}$$

$$= \frac{(-1+\sqrt{3}) + i(1+\sqrt{3})}{2}$$

$$\therefore z = \frac{-1+\sqrt{3}}{2} + i\frac{1+\sqrt{3}}{2}$$

$$r = \sqrt{\left(\frac{-1+\sqrt{3}}{2}\right)^2 + \left(\frac{1+\sqrt{3}}{2}\right)^2}$$

$$r = \sqrt{\frac{8}{4}} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{\frac{1+\sqrt{3}}{2}}{\frac{-1+\sqrt{3}}{2}}\right) = 75° = \frac{5\pi}{12}$$

$$\therefore z = \sqrt{2}\left(\cos\frac{5\pi}{12} + \sin\frac{5\pi}{12}\right)$$

55. (2)

Let pint $P : (h,k)$, M(1,2) and N(–2,1)

$PM + PN = 14$

$\Rightarrow (h-1)^2 + (k-2)^2 + (h+2)^2 + (k-1)^2 = 14$

$\Rightarrow 2h^2 + 2k^2 + 2h - 6k - 4 = 0$

Locus of $P : x^2 + y^2 + x - 3y - 2 = 0 \qquad \ldots.(i)$

$x^2 + x - 2 = 0$

$x = 1, x = -2$

Intersection with y-axis, put $x = 0$ in eqn. (i)

$y^2 - 3y - 2 = 0$

$\Rightarrow y = \frac{3 \pm \sqrt{17}}{2}$

Area of the quadrilateral $ACBD$ is

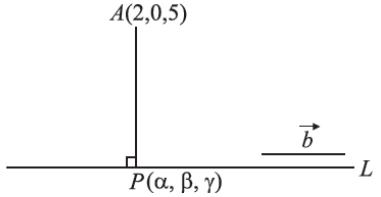
$$= \frac{1}{2}\left(|x_1| + |x_2|\right)\left(|y_1| + |y_2|\right)$$

$$\left[\because \text{ Area } = \frac{1}{2} d_1 d_2 \text{ where } d_1, d_2 \text{ are diagonals }\right]$$

$$= \frac{1}{2} \times 3 \times \sqrt{17} = \frac{3\sqrt{17}}{2}$$

56. (3)

$$L : \frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1} = \lambda \text{ (let)}$$



Let foot of perpendicular is

$P(2\lambda-1,5\lambda+1,-\lambda-1)$

$\overrightarrow{PA}=(3-2\lambda)\hat{i}-(5\lambda+1)\hat{j}+(6+\lambda)\hat{k}$

Direction ratio of line $\Rightarrow \vec{b}=2\hat{i}+5\hat{j}-\hat{k}$

Now, $\Rightarrow \overrightarrow{PA}\cdot\vec{b}=0$

$\Rightarrow 2(3-2\lambda)-5(5\lambda+1)-(6+\lambda)=0$

$\Rightarrow \lambda=\frac{-1}{6}$

$P(2\lambda-1,5\lambda+1,-\lambda-1)\equiv P(\alpha,\beta,\gamma)$

$\Rightarrow \alpha=2\left(-\frac{1}{6}\right)-1=-\frac{4}{3}\Rightarrow \alpha=-\frac{4}{3}$

$\Rightarrow \beta=5\left(-\frac{1}{6}\right)+1=\frac{1}{6}\Rightarrow \beta=\frac{1}{6}$

$\Rightarrow \gamma=-\lambda-1=\frac{1}{6}-1\Rightarrow \gamma=-\frac{5}{6}$

So, $\frac{\beta}{\gamma}=\frac{-1}{5}\neq -5$

57. (1)

$\sum f_i=62=3k^2+4k-2=62$

$\Rightarrow 3k^2+16k-12k-64=0$

$\Rightarrow k(3k+16)-4(3k+16)=0$

$k=4$ or $-\frac{16}{3}$(rejected)

$\mu=\frac{\sum f_i x_i}{\sum f_i}$

$\mu=\frac{8+2(15)+3(15)+4(17)+5}{62}=\frac{156}{62}$

Variance $=\sigma^2=\frac{\sum f_i x_i^2}{\sum f_i}-\mu$

$=\frac{8\times1^2+15\times13+17\times16+25}{62}-\left(\frac{156}{62}\right)^2$

$\sigma^2=\frac{500}{62}-\left(\frac{156}{62}\right)^2$

$\Rightarrow \sigma^2+\mu^2=\frac{500}{62}=8.064\left[\because \mu=\frac{156}{62}\right]$

Hence, $\left[\sigma^2+\mu^2\right]=8$

58. (2)

$S=3+4+8+9+\underbrace{13+14}+\underbrace{18+19}\ldots..40$ terms

$\Rightarrow S=7+17+27+37+47+\ldots 20$ terms

$\Rightarrow S_{40}=\frac{20}{2}[2\times7+(19)10]=10[14+190]$

$=[2040]=(102)(20)\Rightarrow m=20$

59. (2)

Focus and vertex of parabola $y^2 = 2x$ is (1/2, 0) and (0,0) respectively.

Let the centre of the circle having radius 2 units be (h, k).

$\therefore$ Its equation will be $(x-h)^2+(y-k)^2=4$

The circle passes through the pints (0, 0) and (1/2, 0).

$\therefore (1/2-h)^2+k^2=h^2+k^2=4\Rightarrow h=1/4$

$\therefore 1/16+k^2=4\Rightarrow k=\pm\frac{\sqrt{63}}{4}$

So, possible equation of circles are

$\left(x-\frac{1}{4}\right)^2+\left(y\pm\frac{\sqrt{63}}{4}\right)^2=4$

If parabola $y=\left(x-\frac{1}{4}\right)^2+\alpha$ and circle

$\left(x-\frac{1}{4}\right)^2+\left(y-\frac{\sqrt{63}}{4}\right)^2=4$ touch then

$\alpha=2+\frac{\sqrt{63}}{4}=\frac{8+\sqrt{63}}{4}.$

If parabola $y=\left(x-\frac{1}{4}\right)^2+\alpha$ and circle

$\left(x-\frac{1}{4}\right)^2+\left(y+\frac{\sqrt{63}}{4}\right)^2=4$ touch then

$\alpha=2-\frac{\sqrt{63}}{4}=\frac{8-\sqrt{63}}{4}.$

$\therefore 4\alpha-8=\pm\sqrt{63}$ or $(4\alpha-8)^2=63$

60. (1)

$|2x + 3|$ is not differentiable at $x = -3/2$

$9\left[x+\frac{1}{2}\right]$ is not differentiable when

$x+\frac{1}{2}=k,$ where k is an integer

or $x=k-\frac{1}{2}$, where $-20<k-\frac{1}{2}<20$ or $-\frac{39}{2}<k<\frac{41}{2}$

$\therefore k=-19,-18,-17,\ldots\ldots,18,19,20$

$\therefore$ So, there will be 39 new points of non-differentiability as $x = -3/2$ is already counted.

$[x + 10]$ or $[x] + 10$ will be non-differentiable at $x = -19, -18, -17, \ldots, 18, 19$.

So, there will be 39 more points of non-differentiability. So, total number of points of non-differentiability = 1+ 39 +39 = 79

61 **(1)**

For number to be divisible by 15, last digit should be 5 and sum of digits must be divisible by 3.

1	2	1	5

Numbers = 3

2	2	3	5

Numbers = 3

3	3	1	5

Numbers = 3

1	1	5	5

Numbers = 3

2	3	5	5

Numbers = 6

3	5	5	5

Numbers = 3

Total Numbers = 21

62. **(3)**

$$y = \text{cosec}\left(2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right)$$

Let $\cot^{-1}(5) = \theta$ and $\cos^{-1}\left(\frac{4}{5}\right) = \alpha$

$$\Rightarrow \cot\theta = 5\cos\alpha = \frac{4}{5}$$

$$y = \text{cosec}(2\theta + \alpha) = \frac{1}{\sin(2\theta + \alpha)}$$

$$= \frac{1}{\sin 2\theta\cos\alpha + \cos 2\theta\sin\alpha} = \frac{1}{\frac{5}{13}\cdot\frac{4}{5} + \frac{12}{13}\cdot\frac{3}{5}}$$

$$= \frac{65}{20+36} = \frac{65}{56}$$

as $\begin{cases} \sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta} = \dfrac{2\left(\frac{1}{5}\right)}{1+\frac{1}{25}} = \dfrac{5}{13} \\ \cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta} = \dfrac{1-\frac{1}{25}}{1+\frac{1}{25}} = \dfrac{12}{13} \end{cases}$

63. **(4)**

A = {10, 9, 8,7, 6, 4, 3, 0}

R = {(10, 9), (10,8), (10,7), (10,3), (9,8), (9,7), (9,6), (9,4), (9,0), (8,7), (8,6), (8,3), (7,6), (7,4), (7,0), (6, 4), (6,3) ,(4,3), (3,0)}

All the elements of R, (a, b) are of $a > b$. Hence, we need to add total of 19 more elements to R to make in symmetric.

64. **(3)**

$$\tan^{-1}\left[\frac{2x+3x}{1-6x^2}\right] = \frac{\pi}{4}$$

$$\Rightarrow 1-6x^2 = 5x$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow x = \frac{1}{6}, -1$$

Hence, $x = 1/6$ $\quad [\because x > 0]$

65. **(1)**

$$\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}$$

$$\vec{b} = \hat{i} + \hat{k}$$

$$\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{r}\times\vec{a} = \vec{c}\times\vec{a} \Rightarrow (\vec{r} - \vec{c})\times\vec{a} = \vec{0}$$

$\Rightarrow \vec{r} - \vec{c}$ is parallel to $\vec{a}$

$$\therefore \vec{r} = \vec{c} + \lambda\vec{a}$$

$$\vec{r}\cdot\vec{b} = 0 \Rightarrow \vec{c}\cdot\vec{b} + \lambda\vec{b}\cdot\vec{a} = 0$$

$[\because \vec{b}\cdot\vec{a} = 2+5 = 7, \vec{b}\cdot\vec{c} = -2]$

$$\therefore \vec{r} = \vec{c} + \frac{2\vec{a}}{7} = \frac{1}{7}(11\hat{i} - 11\hat{k})$$

$$|\vec{r}| = \frac{11\sqrt{2}}{7}$$

66. **(2)**

Given $2a + 2b = 4(2\sqrt{2} + \sqrt{14})$

$$2a + 2b = 4\sqrt{2}(2 + \sqrt{7}) \qquad \ldots.(i)$$

$$b^2 = a^2\left(\frac{11}{4} - 1\right)$$

$$4b^2 = 7a^2$$

$$b = \frac{\sqrt{7}}{2}a \qquad \ldots.(ii)$$

One solving equation (i) and equation (ii), we get

$$a = 4\sqrt{2}, b = 2\sqrt{14}$$

Hence, $a^2 + b^2 = 88$

67. **(2)**

We have, $\frac{dy}{dx} = \frac{ax - by + a}{bx + cy + a}$

$$= bxdy + cydy + ady = axdx - bydx + adxn$$

$$= cydy + ady - axdx - adx + b(xdy + ydx) = 0$$

$$= c\int ydy + a\int dy - a\int xdx - a\int dx + b\int d(xy) = \int 0$$

$$= \frac{cy^2}{2} + ay - \frac{ax^2}{2} - ax + bxy = k$$

$$= ax^2 - cy^2 + 2ax - 2ay - 2bxy = k_1$$

Since, the above equation is of a circle

$\Rightarrow a = -c$ and b = 0

$ax^2 + ay^2 + 2ax - 2ay = k_1$

$\Rightarrow x^2 + y^2 + 2x - 2y = \lambda$

$\Rightarrow \lambda = \frac{k_1}{a}\Big]$

Since, the given circle passes through (2,5)

$4 + 25 + 4 - 10 = \lambda \Rightarrow \lambda = 23$

Circle $= x^2 + y^2 + 2x - 2y - 23 = 0$

Centre (–1,1)

$r = \sqrt{(-1)^2 + 1^2 + 23} = 5$

Shortest distance of (11, 6)

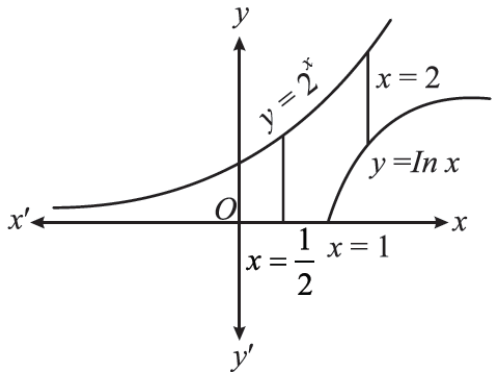
$= \sqrt{12^2 + 5^2} - 5$

$= 13 - 5$

$= 8$

68. (2)

$$R = \left\{(x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2\right\}$$



Area $= \int_{\frac{1}{2}}^{2} 2^x dx - \int_1^2 \ln x dx$

$$= \left[\frac{2^x}{\ln 2}\right]_{1/2}^{2} - [x\ln x - x]_1^2$$

$$= \frac{\left(2^2\right) - 2^{1/2}}{\log_e 2} - (2\ln 2 - 1)$$

$$= \frac{\alpha}{\ln 2} + \beta\log_e 2 + \gamma$$

$\therefore \alpha = 2^2 - \sqrt{2}, \beta = -2, \gamma = 1$

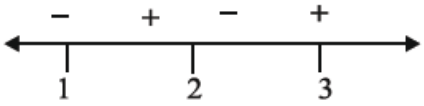
Hence, $(\alpha + \beta - 2\gamma)^2$

$= \left(2^2 - \sqrt{2} - 2 - 2\right)^2$

$= (\sqrt{2})^2 = 2$

69. (4)

$F(x) = (x-1)(x-2)(x-5)$



At $x = 1$ and $x = 5$, $F'(x)$ changes from – to +

Therefore, $F(x)$ has two local minima point at $x = 1$ and $x = 5$

$F(x)$ has one local maxima point at $x = 2$

70. (1)

$5^{99} = 5^4.5^{95}$

$= 625\left[5^5\right]^{19}$

$= 625[3125]^{19}$

$= 625[3124 + 1]^{19}$

$= 625[11k \times 19 + 1]$

$= 625 \times 11k \times 19 + 625$

$= 11k_1 + 616 + 9$

$= 11(k_2) + 9$

Remainder $= 9$

71. (6)

$$I = \int_3^b \frac{1}{3}\left(\frac{1}{x^2-4} - \frac{1}{x^2-1}\right)dx$$

$$\left[\because \frac{1}{(x^2-1)(x^2-4)} = \frac{1}{3}\left\{\frac{1}{x^2-4} - \frac{1}{x^2-1}\right\}\right]$$

$$= \frac{1}{3}\left[\frac{1}{4}\ln\frac{x-2}{x+2} - \frac{1}{2}\ln\frac{x-1}{x+1}\right]_3^b$$

$$= \frac{1}{12}\ln\left(\frac{(b-2)/b+2}{\frac{1}{5}}\right) - \frac{1}{6}\ln\left(\frac{\frac{b-1}{b+1}}{\frac{2}{4}}\right)$$

$$= \ln\left(\left(\frac{5(b-2)}{b+2}\right)^{\frac{1}{12}} / \left(\frac{2(b-1)}{b+1}\right)^{\frac{1}{6}}\right)$$

$$= \ln\left(\frac{49}{40}\right)^{\frac{1}{12}}$$

$$\left[\because 12I = \log_e\left(\frac{49}{40}\right) \Rightarrow I = \log_e\left(\frac{49}{40}\right)^{1/12}\right]$$

$$\Rightarrow \frac{(b-2)(b+1)^2}{(b+2)(b-1)^2} \cdot \frac{5}{4} = \frac{49}{40}$$

$$\Rightarrow \frac{(b-2)(b+1)^2}{(b+2)(b-1)^2}=\frac{49}{50}$$

$$\Rightarrow \frac{(b+1)^2}{(b-1)^2}=\frac{49b+98}{50b-100}$$

$$\Rightarrow \frac{4b}{2\left(b^2+1\right)}=\frac{-b+198}{99b-2}$$

$$\left[\because \frac{a}{b}=\frac{c}{d}\Rightarrow \frac{a-b}{a+b}=\frac{c-d}{c+d}\right]$$

$$\Rightarrow b^3-3b-198=0$$

$$\Rightarrow b=6$$

72. (3)

$S_1: x^2+y^2-x-y-\frac{1}{2}=0$ $C_1:\left(\frac{1}{2},\frac{1}{2}\right)$

$$r_1=\sqrt{\frac{1}{4}+\frac{1}{4}+\frac{1}{2}}=1$$

$S_2; x^2+y^2-4y+\frac{7}{4}=0$ $\quad C_2:(0,2)$

$$r_2=\sqrt{4-\frac{7}{4}}=\frac{3}{2}$$

$S_3: x^2+y^2-4x-2y+5-r^2=0$ $\quad C_3:(2,1)$

$$r_3=\sqrt{4+1-5+r^2}=|r|$$

$$C_1C_3=\sqrt{\frac{5}{2}}, C_2C_3=\sqrt{5}$$

S_1 must be inside $S_3 \therefore \sqrt{\frac{5}{2}}\le |r-1|$

$\Rightarrow r\le 1+\sqrt{\frac{5}{2}}, r\ge \frac{3}{2}+\sqrt{5}$ and also S_2 must be inside

$$S_3 \therefore \sqrt{5}\le \left|r-\frac{3}{2}\right| \Rightarrow r-\frac{3}{2}\ge \sqrt{5}, r-\frac{3}{2}\le -\sqrt{5}$$

73. (12)

$$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}}$$

$$=\int \frac{dx}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^{3/4}(x+2)^{5/4}}$$

$$=\int \frac{dx}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^2}$$

Put $\frac{x-1}{x+2}=t \Rightarrow \frac{3}{(x+2)^2}dx=dt$

$$\therefore \frac{1}{3}\int \frac{dt}{t^{3/4}}=\frac{1}{3}\cdot\frac{t^{1/4}}{\frac{1}{4}}+C=\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$$

74. (2)

$$\frac{5}{216}$$

75. (1)

$$\cos^2 10^\circ-\cos 10^\circ\cos 50^\circ+\cos^2 50^\circ$$

$$=\frac{1}{2}\left(2\cos^2 10^\circ-2\cos 10^\circ\cos 50^\circ+2\cos^2 50^\circ\right)$$

$$=\frac{1}{2}\left(1+\cos 20^\circ-\left(\cos 60^\circ+\cos 40^\circ\right)+1+\cos 100^\circ\right)$$

$$=\frac{1}{2}\left(\frac{3}{2}+\cos 20^\circ+2\sin 70^\circ\sin\left(-30^\circ\right)\right)$$

$$=\frac{1}{2}\left(\frac{3}{2}+\cos 20^\circ-\sin 70^\circ\right)=\frac{3}{4}=l$$

$$=\frac{4l}{3}=1$$



PW Web/App - https://smart.link/7wwosivoicgd4

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